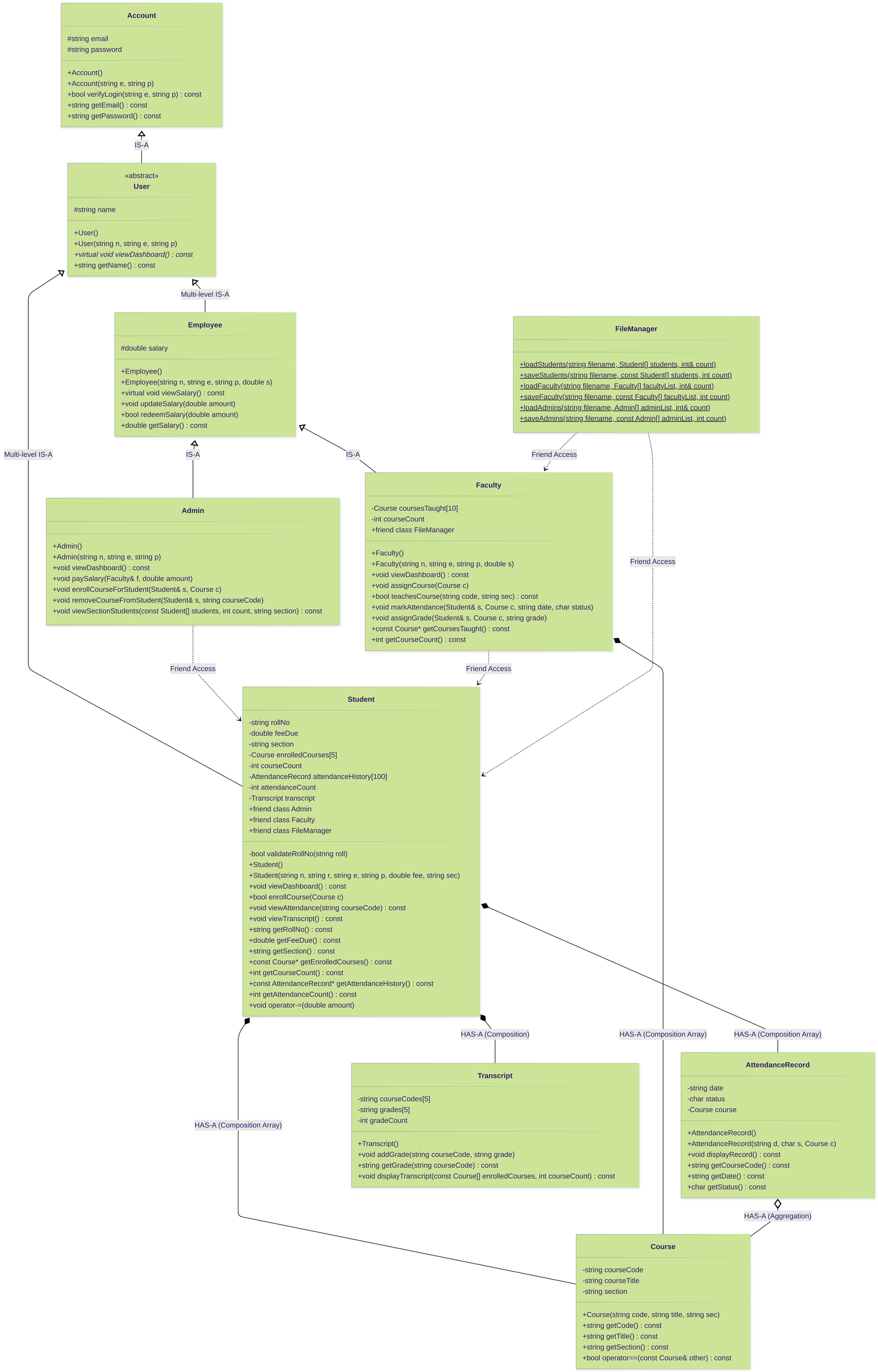
Account
#string email
#string password
+Account()
+Account(string e, string p)
+bool verifyLogin(string e, string p) : const
+string getEmail() : const
+string getPassword() : const
IS-A
«abstract»
User
#string name
+User()
+User(string n, string e, string p)
+virtual void viewDashboard() : const
+string getName() : const
Multi-level IS-A
Employee
#double salary
+Employee()
+Employee(string n, string e, string p, double s)
+virtual void viewSalary() : const
+void updateSalary(double amount)
+bool redeemSalary(double amount)
+double getSalary() : const
FileManager
+loadStudents(string filename, Student[] students, int& count)
+saveStudents(string filename, const Student[] students, int count)
+loadFaculty(string filename, Faculty[] facultyList, int& count)
+saveFaculty(string filename, const Faculty[] facultyList, int count)
+loadAdmins(string filename, Admin[] adminList, int& count)
+saveAdmins(string filename, const Admin[] adminList, int count)
Multi-level IS-A
IS-A
IS-A
Friend Access
Admin
+Admin()
+Admin(string n, string e, string p)
+void viewDashboard() : const
+void paySalary(Faculty& f, double amount)
+void enrollCourseForStudent(Student& s, Course c)
+void removeCourseFromStudent(Student& s, string courseCode)
+void viewSectionStudents(const Student[] students, int count, string section) : const
Faculty
-Course coursesTaught[10]
-int courseCount
+friend class FileManager
+Faculty()
+Faculty(string n, string e, string p, double s)
+void viewDashboard() : const
+void assignCourse(Course c)
+bool teachesCourse(string code, string sec) : const
+void markAttendance(Student& s, Course c, string date, char status)
+void assignGrade(Student& s, Course c, string grade)
+const Course* getCoursesTaught() : const
+int getCourseCount() : const
Friend Access
Friend Access
Friend Access
Friend Access
Student
-string rollNo
-double feeDue
-string section
-Course enrolledCourses[5]
-int courseCount
-AttendanceRecord attendanceHistory[100]
-int attendanceCount
-Transcript transcript
+friend class Admin
+friend class Faculty
+friend class FileManager
-bool validateRollNo(string roll)
+Student()
+Student(string n, string r, string e, string p, double fee, string sec)
+void viewDashboard() : const
+bool enrollCourse(Course c)
+void viewAttendance(string courseCode) : const
+void viewTranscript() : const
+string getRollNo() : const
+double getFeeDue() : const
+string getSection() : const
+const Course* getEnrolledCourses() : const
+int getCourseCount() : const
+const AttendanceRecord* getAttendanceHistory() : const
+int getAttendanceCount() : const
+void operator-=(double amount)
HAS-A (Composition)
HAS-A (Composition Array)
HAS-A (Composition Array)
HAS-A (Composition Array)
Transcript
-string courseCodes[5]
-string grades[5]
-int gradeCount
+Transcript()
+void addGrade(string courseCode, string grade)
+string getGrade(string courseCode) : const
+void displayTranscript(const Course[] enrolledCourses, int courseCount) : const
AttendanceRecord
-string date
-char status
-Course course
+AttendanceRecord()
+AttendanceRecord(string d, char s, Course c)
+void displayRecord() : const
+string getCourseCode() : const
+string getDate() : const
+char getStatus() : const
HAS-A (Aggregation)
Course
-string courseCode
-string courseTitle
-string section
+Course(string code, string title, string sec)
+string getCode() : const
+string getTitle() : const
+string getSection() : const
+bool operator==(const Course& other) : const